



# Xiaofeng Li

AI Undergraduate · LLM Evaluation & Multimodal Perception

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🔗 c-allergic

## Summary

I am Xiaofeng Li, a final-year undergraduate student in Artificial Intelligence at HKUST(GZ), graduating in 2027. My research focuses on large language model evaluation and multimodal perception. I am currently developing HiEval, an agent-based caption evaluation system, and I built IPV-Bench, the first benchmark for image protection methods in image-to-video generation, as first author (under review at AAAI 2027).

## Education

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| <b>BSc</b>              | <b>The Hong Kong University of Science and Technology (Guangzhou),</b> Artificial Intelligence | Guangzhou, China<br>Sept 2023 – June 2027 |
|                         | <ul style="list-style-type: none"><li>GPA: 3.6+/4.3</li><li>Dean's List, Spring 2025</li></ul> |                                           |
| <b>Exchange Student</b> | <b>RWTH Aachen University,</b> Exchange Program                                                | Aachen, Germany<br>Sept 2025 – Feb 2026   |
|                         | <ul style="list-style-type: none"><li>One-semester study exchange in Germany</li></ul>         |                                           |

## Experience

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| <b>Caption Evaluation Agent System (HiEval),</b> Developer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Apr 2026 – present<br>6 months |
| <ul style="list-style-type: none"><li>Built an LLM-agent-based caption evaluation system that hierarchically explores images from coarse to fine (region tree: ground → subdivide → decompose → match → commit → prune), verifying caption completeness against human-perception-aligned criteria and producing 3-D scores (P/R/F1, attributes, relations) with full decision traces — addressing the bias of surface-level element matching in existing benchmarks</li><li>Automated the evaluation pipeline: produces fine-grained evaluation results and improvement suggestions without manual annotation, improving both efficiency and quality</li><li>Overcame self-validation bias by building an independent grounding stack (GroundingDINO + SAM3 for objects, Mask2Former for background/stuff classes) that provides objective visual evidence</li><li>Validated on four public benchmarks (CapsBench, CompreCap, CAPTURE, Perception-Rubrics) against 4 mainstream models (GPT-5.5, Claude Sonnet 4.5, Qwen3.5, Doubao Seed 2.1), evaluating 6600+ captions</li></ul> |                                |

## Publications

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| <b>IPV-Bench: Benchmarking Image Protection Methods under Diverse Image-to-Video Generation Scenarios</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Mar 2026 |
| <p>The first systematic benchmark for image protection methods in image-to-video (I2V) generation scenarios — unifying evaluation protocols and generation settings, covering fidelity-effectiveness trade-offs, cross-model transfer, and robustness across diverse image domains, enabling reproducible and comparable evaluation.</p> <p>Xiaofeng Li, Leyi Sheng, Yifan Zhao, Zhen Sun, Zongmin Zhang, Jiaheng Wei, Xinlei He<br/><a href="https://arxiv.org/abs/2603.26154">arxiv.org/abs/2603.26154</a> (arXiv:2603.26154 (Under review at AAAI 2027))</p> |          |

## Programming Guide for Solving Constraint Satisfaction Problems with Tensor Networks

Jan 2025

A practical programming guide demonstrating how to formulate and solve constraint satisfaction problems (e.g., graph coloring, SAT) with tensor-network-based methods, with reproducible Julia implementations.

Xuanzhao Gao, *Xiaofeng Li*, Jinguo Liu

[arxiv.org/abs/2501.00227](https://arxiv.org/abs/2501.00227) (arXiv:2501.00227)

## Awards

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- Third Prize, Huawei Ascend Operator Challenge (S8), 2026
- Merit Award, Undergraduate Research Project Competition, HKUST(GZ), 2024
- Dean's List, HKUST(GZ), Spring 2025

## Skills

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**Programming Languages:** Python, Shell

**Deep Learning & LLMs:** PyTorch, Transformers, HuggingFace, LLM APIs & inference engineering

**Computer Vision:** GroundingDINO, SAM2/SAM3, Mask2Former

**Tools:** Git/GitHub, Claude Code, Linux, PowerPoint

## Languages

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- Chinese — Native
- English — IELTS 7.5